OUTPUT:

X. Mini Project – Student Academic Performance Analysis

AIM:

To clean, process, and visualize student academic data, including marks, attendance, subject-wise performance, pass percentage, and overall grades using Python. To create static visualizations using Matplotlib and Seaborn, interactive visualizations using Plotly, and dashboards using Power BI and Tableau to evaluate student performance and identify academic trends.

DOMAIN:

Educational Data Analytics

PROBLEM STATEMENT:

In educational institutions, teachers and administrators often find it difficult to analyze students' academic performance across multiple subjects, attendance records, and examinations. Manual analysis takes time and may not clearly identify weak subjects, high-performing students, or the relationship between attendance and marks. A data-driven visualization system is therefore needed to compare subject-wise marks, identify performance trends, evaluate pass percentages, and support better academic decisions.

OUTPUT:

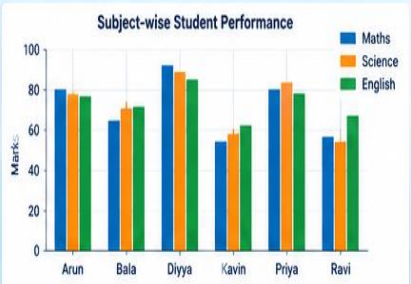
1. Python Output (Matplotlib / Seaborn / Plotly)

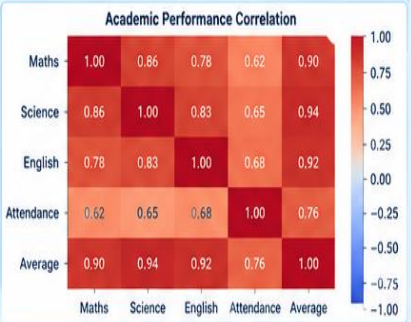
Student Academic Performance

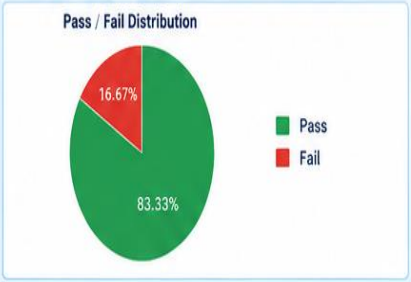
Student	Maths	Science	English	Attendance	Average	Result
Arun	85	78	80	95	81.0	Pass
Bala	65	70	75	80	70.0	Pass
Divya	92	88	90	98	90.0	Pass
Kavin	55	60	58	70	57.7	Pass
Priya	78	82	85	90	81.7	Pass
Ravi	48	45	50	65	47.7	Fail

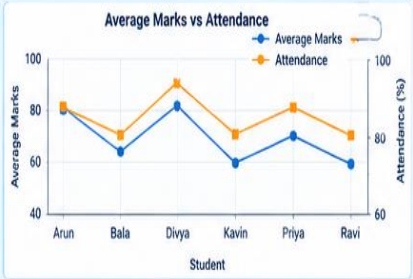
Average Marks by Subject:
Maths 70.5
Science 68.8
English 71.3

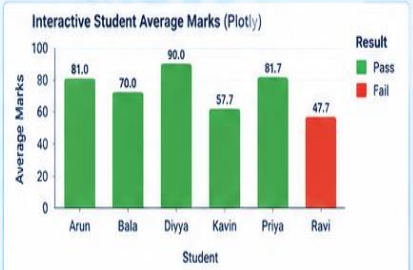
Pass Percentage: 83.33 %









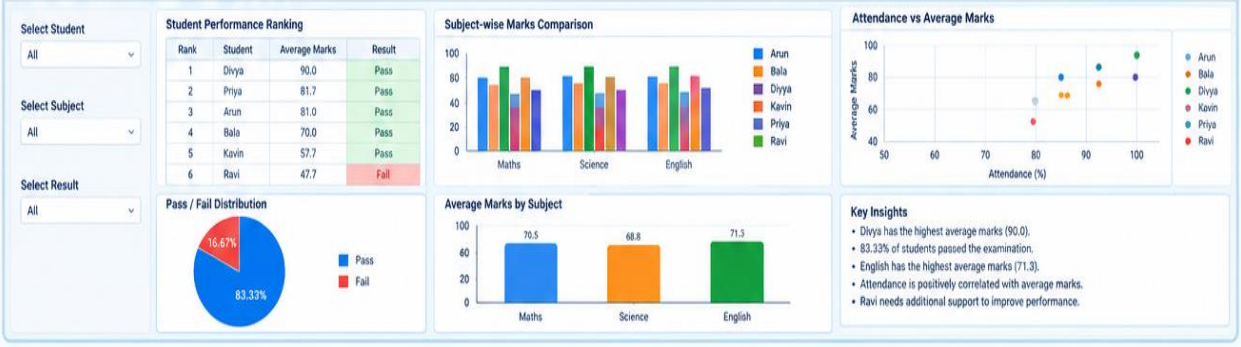


2. Power BI Dashboard Output



Rank	Student	Average Marks	Result
1	Divya	90.0	Pass
2	Arun	81.0	Pass
3	Priya	81.7	Pass
4	Bala	70.0	Pass
5	Kavin	57.7	Pass

3. Tableau Dashboard Output



Rank	Student	Average Marks	Result
1	Divya	90.0	Pass
2	Priya	81.7	Pass
3	Arun	81.0	Pass
4	Bala	70.0	Pass
5	Kavin	57.7	Pass
6	Ravi	47.7	Fail

OUTPUT:

Python:

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import plotly.express as px

data = {
    "Student": ["Arun", "Bala", "Divya", "Kavin", "Priya", "Ravi"],
    "Maths": [85, 65, 92, 55, 78, 48],
    "Science": [78, 70, 88, 60, 82, 45],
    "English": [80, 75, 90, 58, 85, 50],
    "Attendance": [95, 80, 98, 70, 90, 65]
}

df = pd.DataFrame(data).drop_duplicates()
df = df.fillna(df.mean(numeric_only=True))
subjects = ["Maths", "Science", "English"]

df["Average"] = df[subjects].mean(axis=1)
df["Result"] = df["Average"].apply(
    lambda x: "Pass" if x >= 50 else "Fail"
)

print("Student Academic Performance")
print(df)
print("\nAverage Marks by Subject")
print(df[subjects].mean())
print("\nPass Percentage:",
      round((df["Result"] == "Pass").mean() * 100, 2), "%")

df.set_index("Student")[subjects].plot(
    kind="bar", figsize=(9, 4)
)
plt.title("Subject-wise Student Performance")
plt.ylabel("Marks")
plt.xticks(rotation=0)
plt.tight_layout()
plt.show()

sns.heatmap(df[subjects + ["Attendance", "Average"]].corr(),
            annot=True, cmap="coolwarm")
plt.title("Academic Performance Correlation")
plt.tight_layout()
plt.show()

fig = px.bar(df, x="Student", y="Average", color="Result",
             title="Interactive Student Average Marks")
fig.show()
```

Procedure:

1. Collect student marks and attendance data.

2. Load the data into Python using Pandas.

3. Clean missing values and remove duplicate records.

4. Calculate average marks and pass/fail status.

5. Create charts using Matplotlib, Seaborn, and Plotly.

6. Import the dataset into Power BI and Tableau to create dashboards.

OUTPUT:

Power BI:

Create a dashboard showing total students, average marks by subject, student-wise performance, pass/fail percentage, and attendance percentage.

Visual	Purpose
Card visuals	Total students and overall average marks
Bar chart	Compare average marks across subjects
Donut chart	Show pass/fail distribution
Column chart	Compare student-wise performance
KPI / Gauge	Display attendance percentage

Tableau:

Develop an interactive dashboard to rank student performance, compare subject-wise marks, explore attendance versus average marks, and filter results by student or subject.

Worksheet	Visualization
Student Performance	Bar chart of average marks
Subject Comparison	Subject-wise marks chart
Attendance Analysis	Scatter plot of attendance and average marks
Result Summary	Pass/fail distribution

RESULT:

Thus, the Student Academic Performance Analysis mini-project was designed using Python libraries for data processing and visualization, with Power BI and Tableau dashboards for academic analysis. The project helps compare student performance, examine attendance, and identify academic trends.